Causal Inference with LLMs

Day 3: Estimation & Auditing

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instats Seminar

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Day 3: Estimation & Auditing

- Estimation with LLM-Derived Variables
- High-Dimensional Estimation and DML
- Measurement Correction: DSL
- Auditing and Validation

Interactive App

- **URL:** Click here to open the Interactive Streamlit App
- **No Installation:** Runs entirely in the browser via Streamlit.
- **Navigation:** Use the sidebar to switch between workshop days.
- **Step-by-Step:** Each page walks through core computations.
- **Consistency:** Uses the U.S. Congressional speeches dataset (Hein Bound 111); results match local scripts.

From Estimation to Measurement-Aware Inference

Recap:

- Estimation assumes (Y_i, D_i, X_i) are directly observed
- Target estimand:

$$\tau = \mathbb{E}[Y(1) - Y(0)]$$

LLM setting:

- Variables are constructed as *outputs* of the LLM mapping applied to raw data W_i:

$$(\tilde{Y}_i, \tilde{D}_i, \tilde{X}_i) = f_{\mathrm{LLM}}(W_i)$$

- $(\tilde{Y}_i, \tilde{D}_i, \tilde{X}_i)$ are proxies, not the underlying variables

Implication:

- Estimation uses measured variables, not underlying ones
- Introduces bias beyond classical confounding

Target and Induced Estimands

Target:

- $\tau = \mathbb{E}[Y(1) - Y(0)]$

Induced (proxy-based):

- The naive estimator directly uses measured proxies:

$$\tilde{\tau} = \mathbb{E}[\tilde{Y} \mid \tilde{D} = 1] - \mathbb{E}[\tilde{Y} \mid \tilde{D} = 0]$$

- $\tilde{\tau}$ is the effect on *proxies* — it may not identify τ regardless of sample size

Key issue:

- $\tilde{\tau} \neq \tau$ due to measurement distortion
- Identification may fail after measurement

Measurement Error Model

Decomposition:

$$\tilde{Y}_i = Y_i + \varepsilon_i$$

Classical error (MCAR):

- $\varepsilon_i \perp (D_i, X_i)$, so $\mathbb{E}[\varepsilon_i \mid D_i, X_i] = 0$
- Attenuates estimates but does not introduce systematic bias

Non-classical error (LLM setting):

- $\varepsilon_i = \varepsilon(D_i, X_i)$: depends on treatment and latent covariates

$$\mathbb{E}[\varepsilon_i \mid D_i, X_i] \neq 0$$

- Bias varies across groups and persists regardless of sample size

Bias in Treatment Effect Estimation

Observed contrast:

$$\mathbb{E}[\tilde{Y} \mid D = 1] - \mathbb{E}[\tilde{Y} \mid D = 0]$$

Decomposition:

$$= \tau + (\mathbb{E}[\varepsilon \mid D = 1] - \mathbb{E}[\varepsilon \mid D = 0])$$

Conclusion:

- Second term = measurement-induced bias
- Bias persists even without confounding

Text-Induced Confounding

Observation:

- Raw text W_i encodes latent speaker characteristics that may confound the treatment–outcome relationship

LLM mapping:

- $W_i \to (\tilde{D}_i, \tilde{Y}_i, \tilde{X}_i)$

Implication:

- The LLM extracts only a *subset* of the confounding information in W_i
- Residual confounders remain in W_i but are absent from $\tilde{X}_i$
- Ignorability may fail for $(\tilde{Y}, \tilde{D}, \tilde{X})$ even if it holds for (Y, D, W)

High-Dimensional Covariates

Setting:

- X_i may be high-dimensional (embeddings, text features)

Nuisance functions (intermediate objects needed for identification)**:**

$$g(x) = \mathbb{E}[Y \mid X = x], \quad e(x) = P(D = 1 \mid X = x)$$

Goal:

- Estimate causal parameter while controlling for complex X_i

Partially Linear Model

$$Y_i = \theta D_i + g(X_i) + \varepsilon_i$$

Components:

- θ: causal parameter of interest
- $g(X_i)$: unknown confounding function (nonparametric)
- ε_i: idiosyncratic error term

Identifying assumption:

$$\mathbb{E}[\varepsilon_i \mid D_i, X_i] = 0$$

Objective:

- Estimate θ without specifying $g(\cdot)$

Orthogonalization

Nuisance estimates (intermediate objects needed for identification)**:**

- $\hat{g}(X_i)$: estimate of $g(X_i) = \mathbb{E}[Y \mid X_i]$
- $\hat{e}(X_i)$: estimate of $e(X_i) = P(D = 1 \mid X_i)$

Residuals:

$$R_i^Y = Y_i - \hat{g}(X_i), \qquad R_i^D = D_i - \hat{e}(X_i)$$

Resulting model:

$$R_i^Y = \theta R_i^D + \text{noise}$$

Role:

- Removes first-order dependence on nuisance estimation errors

Double Machine Learning: Setup

Objective:

- Estimate θ in the partially linear model:

$$Y_i = \theta D_i + g(X_i) + \varepsilon_i$$

- $g(\cdot)$ is an unknown nonparametric function

Nuisance functions (intermediate objects needed for identification)**:**

- $g(X_i) = \mathbb{E}[Y \mid X_i]$ (outcome regression)
- $e(X_i) = P(D = 1 \mid X_i)$ (propensity score)
- $\hat{g}(X_i), \hat{e}(X_i)$: ML estimates

Double Machine Learning: Orthogonalization

Orthogonalization:

- Construct residuals:

$$R_i^Y = Y_i - \hat{g}(X_i), \quad R_i^D = D_i - \hat{e}(X_i)$$

Estimation step:

- Regress residualized outcome on residualized treatment:

$$R_i^Y = \theta R_i^D + \text{noise} \quad \Rightarrow \quad \hat{\theta} = \frac{\sum_{i=1}^n R_i^D R_i^Y}{\sum_{i=1}^n (R_i^D)^2}$$

Key property:

- Orthogonality $\Rightarrow$ robustness to errors in $\hat{g}, \hat{e}$

Cross-Fitting

Setup:

- Partition sample into K disjoint folds $\mathscr{I}_1, \ldots, \mathscr{I}_K$

Nuisance estimation (intermediate objects needed for identification)**:**

- For each fold k, estimate $\hat{g}^{(-k)}(\cdot)$ and $\hat{e}^{(-k)}(\cdot)$ using all other folds $\mathscr{I}_{-k}$

Residual construction:

- For $i \in \mathscr{I}_k$, compute:

$$R_i^Y = Y_i - \hat{g}^{(-k)}(X_i), \qquad R_i^D = D_i - \hat{e}^{(-k)}(X_i)$$

Role:

- Residuals for unit i use functions estimated without i
- Eliminates overfitting bias, restores valid asymptotics

Design-Based Supervised Learning (DSL)

Definition:

- Uses a small labeled sample $\mathscr{L}$ to learn how to correct the LLM measurement, then applies the correction to the large sample $\mathscr{U}$

Data structure:

- $\mathscr{L}$: small labeled sample with $(Y_i, \tilde{Y}_i, D_i, X_i)$, where Y_i is a human-verified true label
- $\mathscr{U}$: large unlabeled sample with $(\tilde{Y}_i, D_i, X_i)$

Problem:

- True outcomes Y_i observed only in $\mathscr{L}$
- Estimation in $\mathscr{U}$ relies on biased $\tilde{Y}_i$

Key idea:

- Learn correction on $\mathscr{L}$, transfer to $\mathscr{U}$

DSL: Identification and Estimation

Core object:

$$m(x, d, \tilde{y}) = \mathbb{E}[Y \mid X = x, D = d, \tilde{Y} = \tilde{y}]$$

Interpretation:

- Conditional expectation of the true outcome given observed variables
- Captures systematic measurement error in $\tilde{Y}_i$

Identification step:

- Replace missing Y_i in $\mathscr{U}$ by

$$\hat{Y}_i^{\text{corr}} = \hat{m}(X_i, D_i, \tilde{Y}_i)$$

ATE representation:

- Define $m_d(x, \tilde{y}) = \mathbb{E}[Y \mid X = x, D = d, \tilde{Y} = \tilde{y}]$
- Then

$$\tau = \mathbb{E}\big[m(X, 1, \tilde{Y}) - m(X, 0, \tilde{Y})\big]$$

Estimation:

- Estimate $m(\cdot)$ on $\mathscr{L}$ and evaluate on $\mathscr{U}$

DSL vs. Proxy Regression

Common confusion: DSL is not the same as regressing on LLM outputs.

	Naive Regression	**DSL-Corrected**
Outcome used	$\tilde{Y}_i$ (biased proxy)	$\hat{Y}_i^{\text{corr}}$ (corrected)
Labeled data	Not required	Required ($\mathscr{L}$)
Bias	Persists	Reduced
Coefficient	$\tilde{\tau} \neq \tau$	Closer to τ

Key point:

- DSL uses $\mathscr{L}$ to shift the coefficient from $\tilde{\tau}$ toward the true τ
- Without correction, naive regression inherits all measurement bias

Properties and Relation to DML

Double robustness:

- Estimator combines outcome model and weighting model
- Consistent if at least one component is correctly specified

Role in causal pipeline:

- DSL corrects measurement error in outcomes
- DML controls for high-dimensional confounding in X_i

Combined use:

- Apply DSL to construct corrected outcomes
- Apply DML (or IPW / regression) for causal estimation

Prompt Sensitivity

Model:

$$\tilde{Y}_i \sim P(\tilde{Y} \mid W_i, P)$$

Observation:

- Outputs depend on prompt P

Example impact:

- P_1 (Strict): "Is this text explicitly toxic?" $\rightarrow$ Low variance, high precision
- P_2 (Broad): "Is this text somewhat offensive?" $\rightarrow$ High variance, low precision

Implication:

- Measurement becomes stochastic; adds variance and bias

Auditing as a Causal Problem

Objective:

- Assess whether model outputs depend causally on specific inputs
- In particular, detect dependence on sensitive attributes

Key idea:

- Treat the model as a structural mapping

$$\hat{Y}_i = f(W_i)$$

- Study how $\hat{Y}_i$ changes under interventions on components of W_i

Implication:

- Auditing becomes a causal inference problem on the model itself

Interventions on Inputs

Setup:

- W_i: raw input (e.g., text)
- A_i: a specific attribute encoded in W_i (e.g., gender, race)

Intervention:

- Construct a modified input W_i' by changing only A_i
- All other components of W_i are kept fixed

Model outputs:

- $\hat{Y}_i = f(W_i)$: prediction on original input
- $\hat{Y}_i' = f(W_i')$: prediction on modified input

Interpretation:

- Difference $\hat{Y}_i - \hat{Y}_i'$ isolates the effect of A_i on the prediction

Practical note:

- Constructing W_i' requires rewriting the text while preserving all other content — typically via Counterfactual Data Augmentation (CDA) or a separate LLM rewriting step

Individual-Level Causal Effect

Notation:

- $\hat{Y}_i = f(W_i)$: prediction for original input
- $\hat{Y}'_i = f(W'_i)$: prediction after modifying attribute A_i

Definition:

$$\Delta_i = \hat{Y}_i - \hat{Y}'_i$$

Meaning:

- Change in prediction when only A_i is altered
- Measures sensitivity of the model to that attribute

Interpretation:

- $\Delta_i \neq 0$ indicates causal dependence on A_i
- Larger $|\Delta_i|$ indicates stronger local influence

Average Causal Bias

Aggregation:

$$\mathrm{ACB} = \frac{1}{n}\sum_{i=1}^{n}\Delta_i$$

Interpretation:

- Measures systematic effect of A_i across the population
- $\mathrm{ACB} = 0$ suggests no average causal influence

Limitation & Extension:

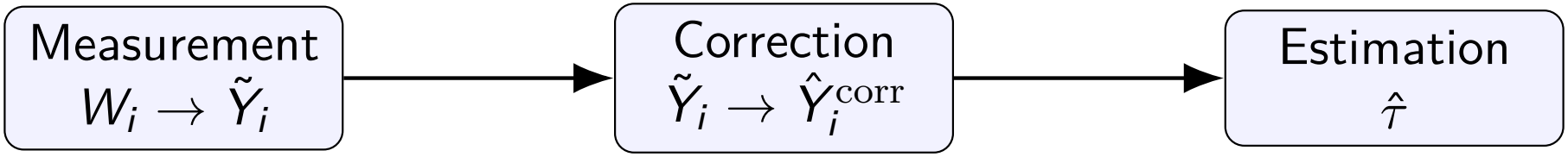
- Zero average does not rule out heterogeneous effects
- Variance $Var(\Delta_i)$ captures dispersion of bias
- Subgroup ACB reveals intersectional bias

Auditing the Full Pipeline

Sources of bias:

- Measurement stage: $W_i \to \tilde{Y}_i$
- Correction stage: $\tilde{Y}_i \to \hat{Y}_i^{\text{corr}}$
- Estimation stage: construction of $\hat{\tau}$

Pipeline view:



Implication:

- Bias may arise and propagate across stages
- Auditing must target the entire pipeline, not only final outputs

Sensitivity to Prompt and Representation

Measurement dependence:

- $\tilde{Y}_i = f_{\mathrm{LLM}}(W_i, P)$ depends on prompt P

Implication:

- Different prompts induce different measurement functions

Example: Effect of Prompt on ATE

Prompt Type	Measurement Error	Estimated ATE
Zero-shot	High variance	Attenuated
Few-shot (balanced)	Low bias	Close to True τ
Chain-of-Thought	Over-confident	Potentially exaggerated*

*Chain-of-Thought prompting may amplify rather than reduce measurement bias; see Egami et al. (2024) for evidence.

Consequence:

- Prompt choice becomes part of the causal design

Key Takeaways

Measurement:

- LLMs introduce non-classical measurement error

Estimation:

- DML controls for high-dimensional confounding
- DSL corrects measurement error using labeled data

Auditing:

- Requires causal interventions on model inputs
- Must be applied to the full pipeline

Overall:

- Measurement, estimation, and validation form a single integrated problem

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Questions and Discussion

Thank you!

Questions?

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